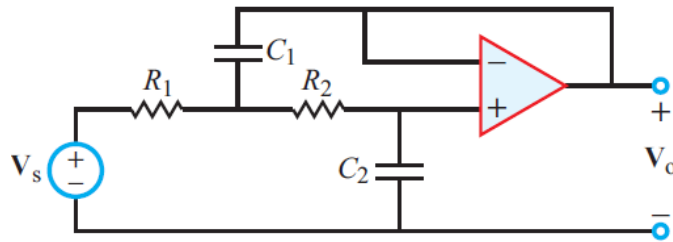


1. (40 points) For the following op-amp circuit,
 - a. (20 points) derive an expression for the frequency response function $H(\omega) = \frac{V_o(\omega)}{V_s(\omega)}$.
 - b. (10 points) What type of filter is it?
 - c. (10 points) What is its maximum gain?



2. (20 Points) A series RLC circuit is driven by an ac source with a phasor voltage $V_s = 10 \angle 30^\circ$ V. If the circuit resonates at 103 rad/s and the average power absorbed by the resistor at resonance is 2.5W, determine the values of R , L , and C given that $Q = 5$.

- 3.(40 points) Derive the impulse response of a system characterized by the frequency response shown in figure below. Express your answer in terms of three sinc functions.

